

**United States Department of Agriculture
Justification for Other than Full and Open Competition
Identification Number: (Solicitation #12639526Q0080)**

Pursuant to the requirements of the Competition in Contracting Act as implemented by Revolutionary Federal Acquisition Regulation (FAR) Overhaul (RFO) FAR Part 6 and in accordance with the requirements of RFO FAR 6.104-1, the justification for the use of the statutory authority cited herein is supported by the following facts and rationale:

(1) Mission Area and Contracting Activity.

The United States Department of Agriculture (USDA), Animal and Plant Health Inspection Service (APHIS), proposes to enter into a contract on a basis other than full and open competition.

(2) Nature and/or description of the action being approved:

This Justification and Approval (J&A) authorizes and approves the award of a sole-source, firm-fixed price, contract for the procurement of J.L. Shepherd 484C-P, Sub-Model 484R-2D, dual chambered, self-shielded, Cobalt-60 irradiators, with optional noiseless air compressor, and associated maintenance and Cobalt reloads for use at the New World Screwworm (NWS) sterilization facility being planned for construction at Moore Air Base (MAB), Edinburg, TX. This procurement is essential for the mass production and targeted dispersal of sterile flies to support the Secretary of Agriculture's five-pronged plan of the NWS Readiness and Response Policy Initiative.

This requirement will be procured utilizing the procedures listed in FAR Part 12.

The identified contractor is:

J.L. Shepherd and Associates
1010 Arroyo St.
818-898-2361
San Fernando, CA 91340
UEI: RN3DK59DTFE7

Actively registered in SAM.gov as a Woman-Owned Small Business (WOSB) as of October 7, 2025, registration date.

USDA has received priority rating authority under the Defense Priorities and Allocations System (DPAS) regulation (15 C.F.R. part 700) for contracts and orders for the construction and operation of the New World Screwworm (NWS) Sterile Fly Production Facility, and is authorized to use the "DX-H8" priority rating on its contracts and orders for all industrial resources needed for the construction and operation of the NWS Sterile Fly Facility, including facility construction and construction materials, production and construction equipment as defined in the DPAS regulation, and items to be used for the manufacturing of production equipment.

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(3) A description of the supplies or services required to meet the agency's needs (including the estimated value).

This contract will provide irradiation capability in support of the NWS Domestic Response. The primary objective of this contract is to ensure adequate, proven and tested, irradiator capability to sterilize 100 million NWS pupae/week within 20 months (November 2027), and 300 million pupae/week within 31 months (December 2028) from the award of the facility construction contract. Pupae sterilization plays a crucial role in the eradication strategy as identified as the third main pillar of the Secretary of Agriculture, "the use of proven sterile insect technology and exploration of additional innovation strategies to defeat the pest."

In addition to procuring irradiators (two initially with an option of two additional units), this contract will include maintenance services for the irradiators and Cobalt-60 reloads that are vital for the continuous operation of the sterilization process. The contracted maintenance services include regular inspections and incidental repairs of safety, electrical, and mechanical systems, and the verification of external radiation levels. Maintenance service aims to optimize equipment performance, enhance operational reliability, and contribute to overall safety. This ongoing support ensures that irradiators remain in peak condition, thereby reducing downtime and associated costs while meeting the sterilization objectives established by the Secretary of Agriculture.

Contracted Cobalt reload service includes the replacement of Cobalt sources once they are depleted from use. Replacing the Cobalt will restore each irradiator's ability to continue providing radiation to sterilize pupae.

It is anticipated that one (1) contract consisting of four (4) base items and twenty (20) optional items will be awarded. The preventive maintenance plans supporting the initial irradiator installations have been prorated in accordance with the planned installation schedule in Calendar Year 2027, preventive maintenance plans cover a full 12-months for all irradiators thereafter. The estimated total for the base year plus all options is as shown below:

Item	Description	Quantity	Unit of Issue
0001	J.L. Shepherd 484C-P Irradiator, Sub-Model 484R-2D w/noiseless air compressor	1	Each
0002	J.L. Shepherd 484C-P Irradiator, Sub-Model 484R-2D w/noiseless air compressor	1	Each
0003	Year 1 Irradiator Preventive Maintenance Plan (for L/I 0001)	1	Yr
0004	Year 1 Irradiator Preventive Maintenance Plan (for L/I 0002)	1	Yr
1001	J.L. Shepherd 484C-P Irradiator, Sub-Model 484R-2D w/noiseless air compressor	1	Each

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Item	Description	Quantity	Unit of Issue
1002	J.L. Shepherd 484C-P Irradiator, Sub-Model 484R-2D w/noiseless air compressor	1	Each
1003	Year 1 Irradiator Preventive Maintenance Plan (for L/I 1001)	1	Yr
1004	Year 1 Irradiator Preventive Maintenance Plan (for L/I 1002)	1	Yr
2001	Year 2 Irradiator Preventive Maintenance Plan (for L/I 0001)	1	Yr
2002	Year 2 Irradiator Preventive Maintenance Plan (for L/I 0002)	1	Yr
2003	Year 2 Irradiator Preventive Maintenance Plan (for L/I 1001)	1	Yr
2004	Year 2 Irradiator Preventive Maintenance Plan (for L/I 1002)	1	Yr
3001	Year 3 Irradiator Preventive Maintenance Plan (for L/I 0001)	1	Yr
3002	Year 3 Irradiator Preventive Maintenance Plan (for L/I 0002)	1	Yr
3003	Year 3 Irradiator Preventive Maintenance Plan (for L/I 1001)	1	Yr
3004	Year 3 Irradiator Preventive Maintenance Plan (for L/I 1002)	1	Yr
4001	Year 4 Irradiator Preventive Maintenance Plan (for L/I 0001)	1	Yr
4002	Year 4 Irradiator Preventive Maintenance Plan (for L/I 0002)	1	Yr
4003	Year 4 Irradiator Preventive Maintenance Plan (for L/I 1001)	1	Yr
4004	Year 4 Irradiator Preventive Maintenance Plan (for L/I 1002)	1	Yr
4005	Cobalt-60 Reload (for L/I 0001)	1	Each
4006	Cobalt-60 Reload (for L/I 0002)	1	Each

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Item	Description	Quantity	Unit of Issue
4007	Cobalt-60 Reload (for L/I 1001)	1	Each
4008	Cobalt-60 Reload (for L/I 1002)	1	Each
			Total

(4) Identify the statutory authority permitting other than full and open competition.

The statutory authority permitting other than full and open competition is 41 U.S.C. 1901 and 1903 as implemented by FAR 12.102.

(5) Demonstrate that the proposed contractor's unique qualifications or the nature of the acquisition requires using the authority cited.

J.L. Shepherd is the original manufacturer of self-shielded irradiators. They provide a complete solution to the irradiator requirement, offering irradiator manufacturing, installation, relocation, maintenance, source reloading, and disposal. J.L. Shepherd is the sole manufacturer of the identical model e.g., model 484C-P, Sub-Model 484R-2D, of NWS irradiators in use at the Panama-United States Commission for the Eradication and Prevention of Screwworm Infestation in Livestock (COPEG) facility located in Panama. This model is currently the only proven and tested model for sterilizing NWS by USDA and thus is the model that will need to be used at the new MAB site's design and operational plan. The specifications and supporting materials and supplies for this specific model is what has been incorporated into the new MAB site's procurement requirements. Utilizing this model achieves two major objectives:

- Utilizes irradiators that have been tested and utilized for the precise purpose of NWS pupae sterilization, in the volume and quantity required. This is in compliance with the Secretary of Agriculture's strategy "use of proven sterile insect technology" and supported but USDA's Office of Homeland Security, Radiation Safety Division; and
- Incorporates a system accounted for in the design and operation of the building being designed at MAB in Edinburg, TX.

Delay in the award of this contract will seriously impact the ability of USDA to meet the Secretary of Agriculture's Readiness and Response Policy Initiative, requiring this facility, and these irradiators, to be operational in 20 months. This facility will be a front-line defense, keeping the NWS from moving to the United States and infecting the Texas cattle overall agricultural industry. As of January 2025, Texas is estimated to have approximately 12.2 million head of cattle, the leading state in beef production in the U.S. The industry contributes approximately \$10 billion in revenues annually. The next potential impact, New Mexico, has an estimated 9,000 beef cattle ranches and contributes over 16,000 jobs, generating \$1.68 billion in output.

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(6) Provide a description of efforts made to ensure that offers are solicited from as many potential sources as possible, including whether a notice was or will be publicized as required by Part 5 and, if not, which exception under Part 5 applies.

Due to the nature of the requirement and the need for the tested and proven irradiator model, as well as the unusual and compelling urgency, no additional sources are planned to be solicited for this effort. When considering possible companies, USDA technical experts took into consideration the required specifications, historical procurement experience for similar purchases, known companies to be licensed for and capable of providing this type of equipment and service, and current use of equipment utilized for the NWS effort. Additionally, internet research was conducted, as described in section 8, to identify potential sources.

While a sources sought notice was not posted for this requirement, the USDA anticipates competing similar procurements for the NWS in the future, as described in section 11.

(7) A determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

To determine fair and reasonable pricing, the USDA reviewed historical pricing data on the USA Spending website, and internal USDA documents, as available, to compare pricing offered to that previously paid. Factors such as inflation, size, and capacity were taken into consideration. From this review, USDA determines that pricing offered for this procurement is expected to be similar to previous pricing paid by USDA, and other Departments, for purchases similar in nature. Specific procurements considered were:

- 89233118CNA000001, FY18 DOE purchase of a self-contained irradiator
- FA820724F0016, FY24 DOD procurement of Cobalt-60 irradiator replacement, decommissioning of existing irradiators.
- N0017306C2090, FY08 DOD procurement of cobalt irradiator.
- W911S609P0223, FY09 Army procurement for a 484R irradiator.
- HHSN272201701336P, FY17 HHS procurement for the reload of 484-R-2 Irradiators.
- 75N93023P01321, FY23 HHS procurement for source exchange services for two 484R irradiators.

(8) A description of the market research conducted, and the results or a statement of the reason Market Research was not conducted.

Extensive market research was conducted to identify alternate sources and potential substitutes that could meet the government's requirements with only minor modifications. USAspending.gov was searched using NAICS 334517, Irradiation Apparatus Manufacturing. The majority of results were related to x-ray scanning machines utilized for security checks. The search field was then narrowed to include PSC 6640, which provided 389 results. A significant number of these results included bone, Computed Tomography (CT), and other types of imaging machines. Award descriptions were reviewed and procurements containing "irradiator" were identified. Of the awards that appeared to be of similar description to this requirement, awards to J.L. Shepherd and Hopewell Designs were found.

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Technical personnel referenced the company Foss Therapy Services in their research. Foss was not included in the findings for NAICS 334517 with PSC 6640, so a search was conducted specifically for Foss. While Foss did have some NAICS 334517 awards, only one was identified as equipment, which was an imaging machine for the National Veterinary Services Laboratory. The other 39 awards identified for Foss had descriptions for repair, maintenance and decommissioning.

It was determined that only J.L. Shepherd was able to manufacture the 484C-P, Sub-Model 484R-2D irradiator model.

GSA/UNICOR/SAM

GSA Advantage and UNICOR searches returned no results for irradiator equipment. GSA returned results for only irradiated supplies.

SAM.gov searches identified some past postings for irradiators, more commonly for maintenance.

No existing IDV contracts for equipment purchases were identified.

Internet

Internet searches were performed using a variety of keywords, including irradiator, Sterile Insect Technique, self-shielded, dual chamber, and “x-ray or gamma ray irradiator”. While initial research shows that x-ray irradiators may be as effective as gamma ray, research papers concurred with USDA technical experts that this technology is not currently available to support the production rates needed for the NWS program.

In searching for gamma ray irradiator providers, three companies were found that provide equipment and services. The findings are as follows:

J.L. Shepherd: Manufacturer of the system used at COPEG e.g., 484C-P, Sub-Model 484R-2D. Original developer of self-shielded irradiators.

Foss Therapy: Lists the 484F on their website as available equipment. From the Registry of Radioactive Sealed Sources and Devices Safety Evaluation of Device document, dated February 12, 2018, it appears that Foss modifies the J.L. Shepherd Model 484 Gamma Irradiator to produce the 484F. Detailed conversations, both verbal and by email, were conducted with Foss Therapy to better understand their capabilities. Foss Therapy stated that they did not have a 484C-P available, and that they will update us if they are able to provide that system, though not very probable. Foss Therapy offered, instead, a single chamber irradiator (812L) and proposed a larger sample cylinder that would hold more pupae than the two cylinders combined in the 484 model. COPEG personnel suggested that the additional volume, in one cylinder, may have an impact on pupae survivability during irradiation. They additionally stated that the cylinder sizes in use at COPEG were specifically determined suitable for NWS irradiation, and that additional testing on larger volume cylinders would not be possible within the timeline required.

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Hopewell Designs: Hopewell Designs advertises a self-contained, single well, gamma ray irradiator. Hopewell does not list a dual chamber irradiator to the size required for the NWS production requirements.

(9) Any other facts supporting the use of other than full and open competition, such as:

Secretary of Agriculture – The Secretary of Agriculture has established an aggressive, five-pronged plan to combat the NWS infestation. This strategy includes the completion of the production facility at Moore Air Base, including the installation of these irradiators, within 20 months for the production of 100 million sterilized pupae/week (November 2027) for Initial Operating Capability (IOC), then stage two, at Full Operating Capability (FOC), for the production of 300 million sterilized pupae/week (early December 2028).

Time – Timeframe for irradiators to be fully operational, based on USDA experience procuring similar irradiators in the past, is 12 months. This includes the manufacturing of the irradiator, the delivery/installation of the irradiators, the manufacturing of the Cobalt-60, the installation of the Cobalt-60 capsules in the irradiators and finally the preparation to begin the sterilization process. Because of shipping packaging requirements, there is an anticipated three-month period between the shipping of each individual irradiator.

Economic/Societal Impacts – From the USDA news release, “New World Screwworm (NWS) is a devastating pest. When NWS fly larvae (maggots) burrow into the flesh of a living animal, they cause serious, often deadly damage to the animal. NWS can infest livestock, pets, wildlife, occasionally birds, and in rare cases, people. This is a serious concern to the U.S. economy and the U.S. food supply and the United States Department of Agriculture (USDA) is committed to combatting the spread of NWS to protect American agriculture.”

(10) A listing of the sources, if any, that expressed an interest in the acquisition in writing

In conversations conducted for market research, Foss Therapy expressed an interest in this requirement. While Foss Therapy did demonstrate knowledge, experience, and capabilities in the Sterile Insect Technique, they were not able to offer the proven and tested model of irradiator required by USDA and used to sterilize NWS pupae at the quantity and rate required.

No additional entities expressed an interest in this procurement.

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- (11) A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition of the supplies or services required.**

To enhance competition and remove barriers for subsequent acquisitions, the USDA plans to attempt developing non-proprietary technical specifications and updating statements of work related to maintenance and Cobalt reload requirements. Additionally, USDA will conduct regular market research and engage with industry to identify new irradiation sources and emerging technologies for potential use in future requirements of a similar nature.

For future non-emergency requirements, attempts will be made to promote vendor capability development through pre-solicitation briefs and technical workshops, offering performance standards early in the acquisition process.

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(12) Certifications & Approvals

Technical / Requirements Personnel Certification

I certify this requirement meets the Government's minimum need and that the supporting data, which forms a basis for this justification, is complete and accurate.

AMY

SPILLMAN

Digitally signed by AMY
SPILLMAN
Date: 2026.03.23 14:47:33
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Technical/Requirements POC

Date

Contracting Officer's Certification

I certify that the data supporting the recommended use of other than full and open competition accurate and complete to the best of my knowledge and belief.

CRAIG

HARPER

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HARPER
Date: 2026.03.23 13:55:25
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Craig Harper
Contracting Officer

Date

Competition Advocate

I have reviewed this justification and find it to be accurate and complete to the best of my knowledge and belief. Therefore, unless additional approvals are required as prescribed below, as advocate for competition for the procuring activity, I hereby approve this justification. If required, my approval constitutes clearance for submission to the Head of the Contracting Activity (HCA).

CRAIG

HARPER

Digitally signed by
CRAIG HARPER
Date: 2026.03.23
13:55:45 -05'00'

Craig Harper
Contracting Office (COCO)/
Competition Advocate

Date